

ENGINEERING MATHEMATICS-IV
MID SEMESTER EVALUATION SCHEME- MAT 2226

11. (a). Let D = defective product by production unit

By the Law of Total Probability:

$$p(D) = p(D | X)p(X) + p(D | Y)p(Y) + p(D | Z)p(Z) = 0.031. \quad (2 \text{ M})$$

$$\text{Compute } p(Y | D) \text{ using Bayes' Theorem: } p(Y | D) = \frac{p(D|Y)p(Y)}{p(D)} \approx 0.34. \quad (1 \text{ M})$$

- (b). To check independence, we need to verify whether: $P(D) = P(Y)$

We already know: $P(Y) = 0.35, P(D) = 0.031$ (from part (a))

Since $P(D) \neq P(Y)$, the events are not independent. (1 M)

12. Let A = defective item

B = item produced by machine 1

$\bar{B}$ = item produced by machine 2

$$P(B) = \frac{1}{3}, P(\bar{B}) = \frac{2}{3}$$

$$P(A | B) = 0.04, P(A | \bar{B}) = 0.02$$

$$P(A) = p(A | B)p(B) + p(A | \bar{B})p(\bar{B}) = \frac{2}{75}. \quad (2 \text{ M})$$

Let X = Number of defective items found

$$X \sim B\left(10, \frac{2}{75}\right)$$

$$\begin{aligned} P(X = 2) &= 10C_2 \left(\frac{2}{75}\right)^2 \left(\frac{73}{75}\right)^8 \\ &= 0.0257 \end{aligned} \quad (1 \text{ M})$$

13. (a) Marginal pdf of X is $g(x) = \int_0^\infty 6e^{-2x-3y} dy$

$$= 6e^{-2x} \left(\frac{e^{-3y}}{-3}\right)_0^\infty$$

$$g(x) = 2e^{-2x}, x > 0. \quad (1 \text{ M})$$

- (b) Marginal pdf of Y is $h(y) = \int_0^\infty 6e^{-2x-3y} dx$

$$= 6e^{-3y} \left(\frac{e^{-2x}}{-2}\right)_0^\infty$$

$$h(y) = 3e^{-3y}, y > 0. \quad (1 \text{ M})$$

- (c) Conditional pdf of X given Y is $\frac{f(x,y)}{h(y)} = \frac{6e^{-2x-3y}}{3e^{-3y}}, x > 0, y > 0. \quad (1 \text{ M})$

- 14.

X	0	-10	10	20	-20
P(X)	$\frac{18}{35}$	$\frac{8}{35}$	$\frac{8}{35}$	$\frac{1}{70}$	$\frac{1}{70}$

(1 ½ M)

$$E(X) = 0$$

(1/2 M)

$$E(X^2) = \frac{400}{7} = V(X)$$

(1 M)

$$15. \rho_{UV} = \frac{E(UV) - E(U)E(V)}{\sqrt{V(U)V(V)}} = \frac{E((X_1 + X_2)(X_1 - X_3)) - E(X_1 + X_2)E(X_1 - X_3)}{\sqrt{V(X_1 + X_2)V(X_1 - X_3)}} \quad (1/2 \text{ M})$$

$$= \frac{V(X_1)}{\sqrt{(V(X_1)+V(X_2))(V(X_1)+V(X_3))}} \quad (1 \frac{1}{2} \text{ M})$$

$$= \frac{25}{\sqrt{75125}} \quad (1 \text{ M})$$

$$16. E(X) = \int_0^\infty x^2 e^{-\frac{x^2}{2}} dx$$

$$\begin{aligned} \text{Let } \frac{x^2}{2} = t. \text{ Then } \int_0^\infty x^2 e^{-\frac{x^2}{2}} dx &= \sqrt{2} \int_0^\infty e^{-t} t^{\frac{1}{2}} dt \\ &= \sqrt{2} \Gamma\left(\frac{3}{2}\right) \\ &= \sqrt{\frac{\pi}{2}} \end{aligned}$$

(1 ½ M)

$$E(X^2) = \int_0^\infty x^3 e^{-\frac{x^2}{2}} dx$$

$$\text{Let } \frac{x^2}{2} = t. \text{ Then } \int_0^\infty x^2 e^{-\frac{x^2}{2}} dx = 2 \int_0^\infty e^{-t} t dt$$

$$= 2\Gamma(2) = 2 \quad (1 \text{ M})$$

$$V(X) = E(X^2) - (E(X))^2 = 2 - \frac{\pi}{2}. \quad (\frac{1}{2} \text{ M})$$

17. Since the probability is proportional to the number on the face, let the probability of rolling a face i be $P(i) = \frac{i}{S}$, where S is the sum of all face values. The sum of all face values is 21.

a) The odd numbers on the die are 1, 3 and 5. So we sum their probabilities:

$$P(\text{odd}) = P(1) + P(3) + P(5) = \frac{1}{21} + \frac{3}{21} + \frac{5}{21} = \frac{9}{21} \quad 1\text{m}$$

b) $P(\text{odd or composite}) = P(1) + P(3) + P(5) + P(4) + P(6) = \frac{19}{21} \quad 1\text{m}$

$$18. \frac{6}{36} + \frac{30}{36} \frac{30}{36} \frac{6}{36} + \left(\frac{30}{36}\right)^4 \frac{6}{36} + \left(\frac{30}{36}\right)^6 \frac{6}{36} + \dots \quad (1 \frac{1}{2} \text{ M})$$

$$= \frac{\frac{6}{36}}{1 - \left(\frac{30}{36}\right)^2} = \frac{6}{11} \quad (1/2 \text{ M})$$

$$19. \text{As } (A \cap \bar{B}) \cup (A \cap B) = A \text{ and } (A \cap \bar{B}) \cap (A \cap B) = \emptyset,$$

$$P(A) = P(A \cap \bar{B}) + P(A \cap B) \quad (1 \text{ M})$$

$$\begin{aligned} \text{Then } P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\ &= P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(\bar{B}). \end{aligned}$$

(1 M)

Hence A and $\bar{B}$ are independent.